Multicomponent magnetization of the Pearson Formation (Great Slave Supergroup, N.W.T.) and the Coronation loop[1]

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The Pearson Formation is the youngest unit of the Great Slave Supergroup of the Coronation Geosyncline. Study of samples from 22 locations shows that the dominant magnetization has a direction 329°, −08° (α_{95} = 10°) and corresponding pole at 195°, 077°W (A_{95} = 9°). This magnetization is considered to have been acquired at the time the rock was formed about −1800 Ma ago. In addition, there are two, possibly three, overprints. One of the two well-established overprints is attributed to uplift following deep burial at about −1600 Ma. The other has no known explanation. These results, together with others from the Coronation Geosyncline, are used to construct a polar loop (the Coronation loop) that is apparently the paleomagnetic signature of the orogeny that affected the geosyncline. The Coronation loop straddles the boundary between the Aphebian and Paleohelikian, and it is tentatively suggested that the physical boundary between these two eras could be placed between the Great Slave Supergroup and the overlying Et-Then Group.

La formation de Pearson est l'unité la plus jeune du supergroupe de Great Slave dans le géosynclinal de Coronation. L'étude de spécimens provenant de 22 localités montre que l'aimantation dominante a une direction de 329°, −08° (α_{95} = 10°) et un pôle correspondant à 195°, 077° ouest (A_{95} = 9°). On considère que cette aimantation a été acquise au moment où la roche s'est formée il y a − 1800 Ma. De plus, il y a deux, peut-être trois, surimpressions. Une des deux qu'on a bien établies est attribuable au soulèvement qui a suivi l'enfouissement profond il y a − 1600 Ma. On n'a aucune explication pour l'autre. On utilise ces résultats et d'autres du géosynclinal de Coronation pour définir une boucle polaire (la boucle de Coronation) qui serait la signature paléomagnétique de l'orogénèse qui a affecté le géosynclinal. La boucle de Coronation chevauche la limite entre l'Aphébien et le Paléohélikien et on suggère provisoirement que la limite physique entre ces deux ères pourrait se placer entre le supergroupe de Great Slave et le groupe de Et-Then sus-jacent.

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Introduction

The Athapuscow Aulacogen is part of the Coronation Geosyncline that developed in Aphebian time along the western margin of the Slave Structural Province in the northwestern part of the Canadian Shield (Hoffman 1973). Rocks of the aulacogen are exposed in the East Arm of Great Slave Lake, N.W.T. The Great Slave Supergroup comprises about 8000 m of sedimentary and volcanic rocks, and forms the bulk of the fill of the aulacogen. The Great Slave sequence was deposited in Aphebian time between about −1800 and −2200 Ma. It rests unconformably on the Union Island Group, is younger than the Wilson Island Group and is overlain unconformably by the Et-Then Group. The lower part of the sequence is intruded by quartz-diorite laccoliths, and all the rocks are intruded by Mackenzie diabase sills and dikes. The distribution of the Great Slave and Et-Then rocks, and the major Mackenzie sills, is shown in Fig. 1. The stratigraphy and geochronological data are summarized in Table 1.

One of the major problems in paleomagnetic studies is the determination of the ages, both relative and 'absolute', of pole positions. In the Athapuscow Aulacogen there are only a few radiometric ages from the stratigraphic sequence itself, but the relative order of rock units is well-known from super-imposition. The sequence is one of the best in the Canadian Shield for magneto-stratigraphic work. The first unit to be studied in the stratigraphic sequence was the Et-Then Group, which was shown to have a magnetization older than the 1250 Ma old Mackenzie intrusions (Irving *et al.* 1972). Subsequent work on the Kahochella Group was important because it showed that the magnetization history of Great Slave Supergroup is complex. Its magnetization consisted of a pre-

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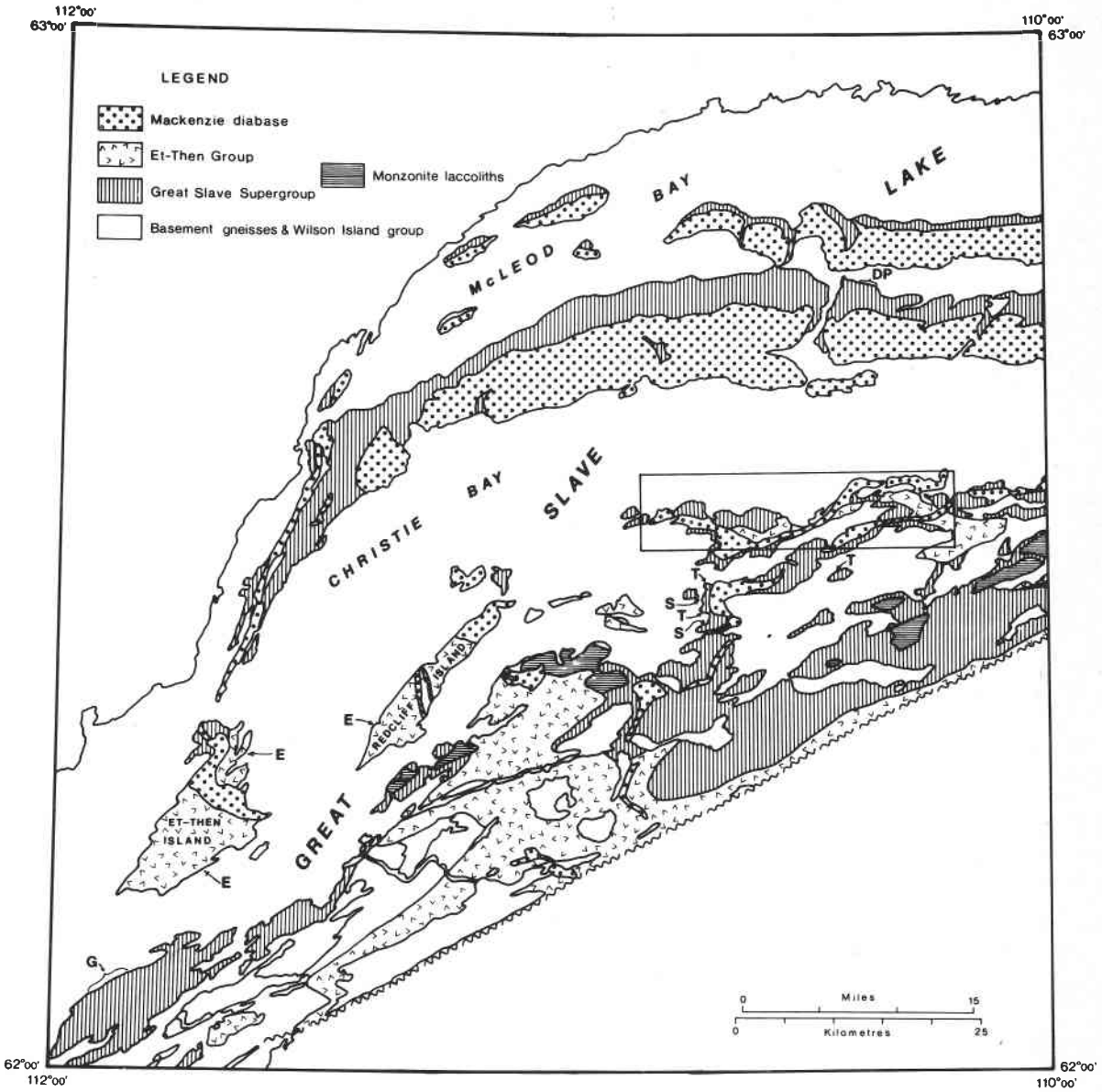


FIG. 1. Geological sketch map of part of the East Arm of Great Slave Lake. The Pearson sampling area is boxed and is shown in more detail in Fig. 2. Sampling areas for other rock units (see Table 1) are as follows: DP, Douglas Peninsular Formation; E, Et-Then Group; G, Gibraltar Formation; S, Stark Formation; T, Tochatwi Formation. The Mackenzie diabase has also been extensively sampled.

folding component on which a stable post-folding component was superimposed (poles UG and KS of Figs. 8, 9, Reid 1972; McMurry *et al.* 1973); although stable secondary magnetizations had been known for many years (Briden 1965) this work was among the first in which stable overprints were quantitatively separated by vector analysis from older, possibly primary magnetizations. More recently, Evans and Bingham (1976) and Bingham and Evans (1976) have shown that complex magnetizations (usually referred to as polyphase or multicomponent magnetizations) are present in the Christie Bay Group. In this paper results are reported from the Pearson Formation, at the top of Great Slave Supergroup, which also has multicomponent magnetizations. This makes a total of 10 paleomagnetically studied units from the Athapuscow Aulacogen (Table 1).

TABLE 1. Table of formations

Mackenzie sheets and dikes (2)(1)(3) −1250 Ma (*a*)
(intrusive contact)
ET-THEN GROUP
Preble Formation (3)
Murky Formation (3)
(unconformity)
Monzonite laccoliths −1800 Ma(*b*)
(intrusive contact)
CHRISTIE BAY GROUP (GS)
Pearson Formation (4)
Portage Inlet Formation
Tochatwi Formation (5)
Stark Formation (6)
(disconformity?)
PETHEI GROUP (GS)
Hearne Formation
Wildbread Formation
Pekanatui Formation
Blanchet Formation
McLean Formation
Utsingi Formation
Taltheilei Formation
Douglas Peninsula Formation (7)
KAHOCHELLA GROUP (GS)
Charlton Bay Formation
McLeod Bay Formation (8)(9)
Gilbraltar Formation (8)(9)
Seton Formation (7) −1870 Ma(*c*)
SOSAN GROUP (GS)
Akaitcho River Formation
Kluziai Formation
(unconformity?)
Duhamel Formation
Hornby Channel Formation
(unconformity?)
UNION ISLAND GROUP
(unconformity?)
Biotite diorite dike −2200 Ma(*d*)
WILSON ISLAND GROUP
(unconformity?)
BASEMENT granodiorite and migmatite

NOTE: Compiled from Hoffman (1968). The four groups comprising the Great Slave Supergroup are labelled GS. Rock units studied paleomagnetically are in italics and numbered as follows: (1) Robertson 1969; (2) Fahrig and Jones 1969; (3) Irving *et al.* 1972; (4) herein; (5) Evans and Bingham 1976; (6) Bingham and Evans 1976; (7) Irving and McGlynn 1978; (8) McMurry *et al.* 1973; (9) Reid 1972. Preliminary results from a few samples from the McLean and Charlton Bay Formations have been given by Reid (1972) but are not noted here. Radiometric ages are as follows: (*a*) K–Ar, Fahrig and Jones 1969; (*b*) K–Ar, Hoffman *et al.* 1974; (*c*) Rb–Sr, Baadsgaard *et al.* 1973; (*d*) K–Ar, Burwash and Baadsgaard 1962. Rb–Sr ages decay constant of 1.39×10^{-11} years^{-1} used throughout.

Geology of the Pearson Formation

The Pearson Formation has a maximum thickness of 165 m and consists of flows of basalt with a few thin beds of grey or red mudstones. It is of limited areal extent, being confined mainly to the region of the inset in Fig. 1. It is autochthonous, being north of the zone of nappes described by Hoffman *et al.* (1977). No radiometric age determinations have been published from the formation. Its stratigraphic position indicates that it is about 1800 Ma old (Table 1). Most flows are massive and have columnar jointing. Many flows are amygdaloidal and have vesicular contacts. The rocks are composed of saussuritized plagioclase lathes in pyroxene that is partly altered to light green actinolite or chlorite. The plagioclase is albite but many lathes have more basic cores. Carbonate occurs along with pumpellyite and the amygdules are filled with quartz and chlorite. Although the presence of prehnite was not confirmed, the mineral assemblage is typical of prehnite–pumpellyite facies metamorphism. This facies is often associated with burial metamorphism and, according to Turner (1968), may indicate a temperature range of 100–250°C and pressures of 2–6 kbars (2×10^2 to 6×10^2 kPa), which correspond to depths of burial of 8–20 km. Liou (1971) suggests a range of 200–350°C and similar pressures, but notes the presence of this assemblage in vertically zoned sequences at depths of 5 km. These depth estimates were made assuming normal thermal gradients, and may be too high because the thermal gradients could have been higher in the past in this deep sedimentary basin. Zen (1974), however, considers these ranges dubious as a general rule and notes that Sigvaldson (1962) has reported pumpellyite in drill holes in the thermal area of Iceland as an alteration product of mafic lavas at depths of only a few hundred metres. Therefore the assemblage in the Pearson lavas may be the result either of deep burial or of deuteric action shortly after the lavas were extruded. Its presence is important in this context, because if it reflects deep burial, then it sets an upper limit to subsequent reheating of the Pearson rocks.

The flows are gently folded about easterly- to northeasterly-trending axes. Dips do not exceed 20° and average about 10°. Samples were collected over about 20 km (Fig. 2). Sampling sites 8–13 and 107–111 are probably from opposing limbs of one fold. The remaining samples are collected from the north limb of a different fold. About five oriented cores were obtained from each site and two specimens cut and studied from each core.

Paleomagnetic Results

The experimental procedures used have been explained previously (Irving and McGlynn 1976*a*). Thermal demagnetization curves were obtained from 23 specimens. There are two distinct types of curves (*A* and *D*), and also intermediate types, which are referred to as *A*+, *B* and *C*.

Specimens with *A* magnetization exhibit ideal behaviour. Their thermal demagnetization curves are square-shouldered, the magnetization first decreasing slowly up to 500°C, and then falling rapidly, all of it being lost at 550°C (Fig. 3). The blocking temperatures are mainly between 500 and

550°C. The carrier of the magnetization is presumably somewhat impure magnetite. The characteristic blocking temperature of titanium-free magnetite (550–570°C) was not observed. After heating to temperatures in the range 300–500°C, the remanence directions are north-northwest with low generally negative inclinations (Fig. 4). Under alternating field demagnetization the *A* magnetization decreases regularly, and the median destructive field (mdf) always exceeds 200 Oe (15.9×10^3 A/m). The directions remain essentially unchanged up to 600 Oe (47.7×10^3 A/m), and in some instances up to 1000 Oe (79.6×10^3 A/m) (Fig. 5). The mean directions at sites with *A* magnetization are shown in Fig. 6.

The *D* phase is the least stable magnetization. It has very low blocking temperatures, generally below 300°C (Fig. 3). The directions are initially very scattered. After cleaning they are directed generally toward the west or southwest, with moderate downward inclination, but the behaviour is far from ideal. The decay of intensity during af (alternating field) demagnetization is very rapid (Fig. 5). Two examples of thermal demagnetization are shown in Fig. 7. In specimen 29 the initial direction is close to the present field and moves to the west between 100 and 400°C, becoming scattered at higher temperatures. In specimen 33 the directions move into the *D* direction at low temperatures, and then at 400 and 500°C move rapidly into the *A* direction. Apparently, in this specimen, the dominant *D* magnetization masks a smaller *A* magnetization; this is the only instance we found of a *D* magnetization occurring with any other magnetization in the same specimen. The mean direction at sites with *D* magnetization is shown in Fig. 6.

B is a subordinate magnetization found at only two sites (42, 43) and for this reason has not been exhaustively studied. Cleaning in alternating fields yields consistent west-northwest directions (Fig. 6), but on thermal demagnetization the directions approach, but never reach, the *C* direction (see below). The magnetization removed during demagnetization (indicated by a diamond in Fig. 4) has a shallow positive inclination.

The *C* magnetization has a thermal decay curve with blocking temperatures distributed mainly between 200 and 500°C (Fig. 3). It is directed to the northwest with intermediate to steep negative inclination (Fig. 6). The *C* magnetization is found in its pure state at only one site (site 40) where the directions are consistent up to 500°C (Fig. 4). Usually it occurs together with *A* magnetization, so that the decay curve is complex. There are blocking temperatures distributed between 100 and 300°C, but between 300 and 500°C there is a distinct shelf;

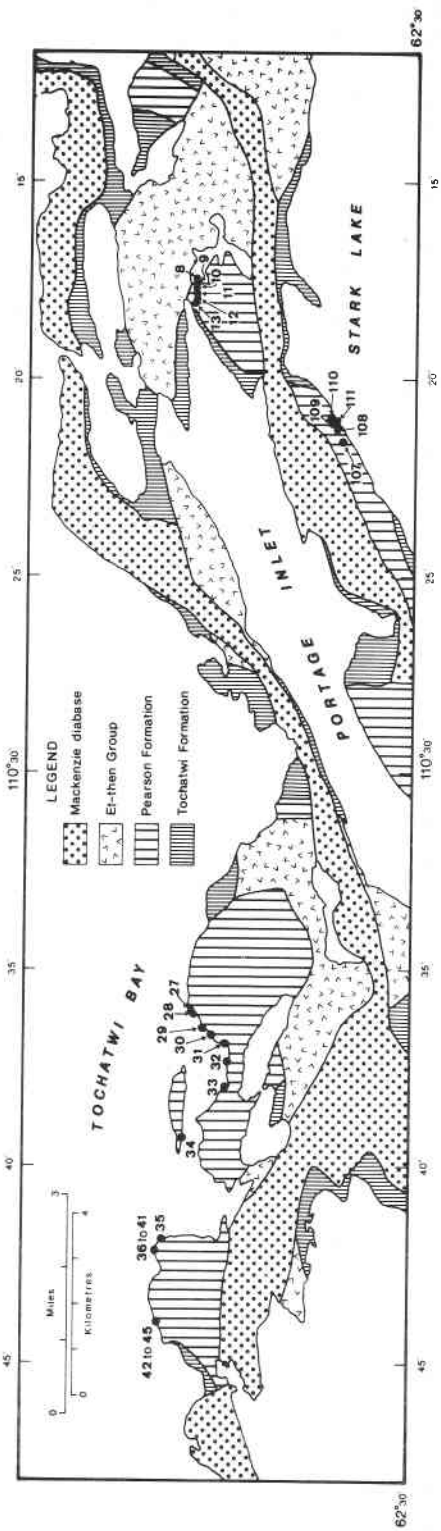


FIG. 2. Sketch map of the sampling area. The sampling sites are numbered. The numbers are laboratory accession numbers and have no stratigraphic significance.

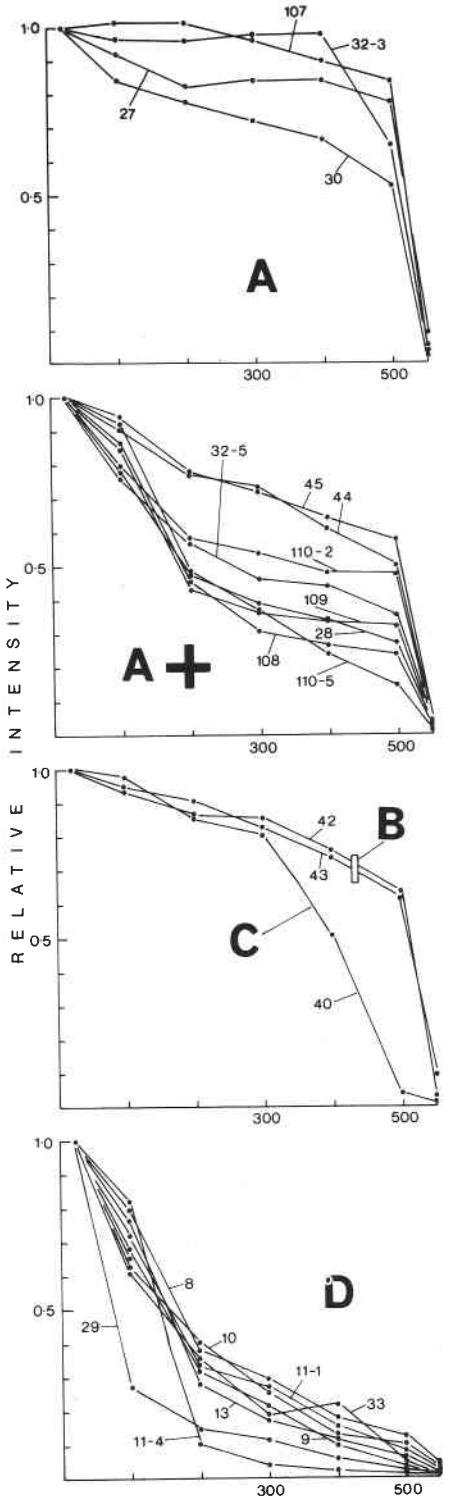


FIG. 3. Thermal demagnetization curves. Relative intensity is plotted as a funcation of temperature (°C). The curves are labelled by sites.

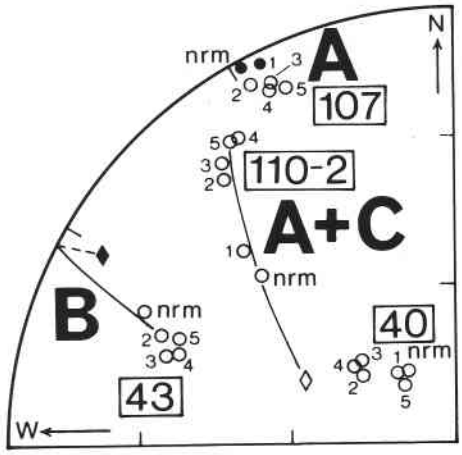


FIG. 4. Direction changes during thermal demagnetization in four specimens labelled by sites. The points are numbers in °C ($\times 10^2$).

the curves have both a slope (*C* magnetization) and a shoulder (*A* magnetization). Such magnetizations are referred to as *A*+. *A*+ magnetization occurs at sites 32, 44, 45, 110 and 111 (Table 2). Typical behaviour of an *A*+ magnetization during a.f. demagnetization is shown in the middle diagram of Fig. 5 (site 110). The initial magnetization is marked nrm (natural remanent magnetization) and is directed toward the north-northwest and upward, but after demagnetization in fields up to 600 Oe (47.7×10^3 A/m) moves to a typical *A* direction with a good end point. In higher fields the directions scatter. The direction of the magnetization obtained by subtracting the end point magnetization from the nrm is marked by *C* in Fig. 5. At sites 44, 45 and 111 the *C* magnetization is much larger than *A*, and the nrm itself is a good estimate of *C*. Typical behaviour during thermal demagnetization is shown in Fig. 4 (specimen 110-2). The behaviour mirrors that just described for af demagnetization.

Samples of interflow baked red mudstone were collected at sites 36–39 and 41, and cherty baked sandstone at site 34. They are intensely fractured and it was difficult to obtain unbroken fresh samples. No coherent magnetizations were observed so no contact test was possible.

Opaque Minerals

The opaque minerals form less than 1% of the rock. Magnetite is the most abundant mineral, and about one-third to one-half of each grain has undergone extensive alteration. In fine-grained lavas the opaque minerals are less than 10 μm in diameter, and difficult to identify. In coarser grained lavas there are grains about 100 μm in diameter, which consist of magnetite with ilmenite lamellae. The

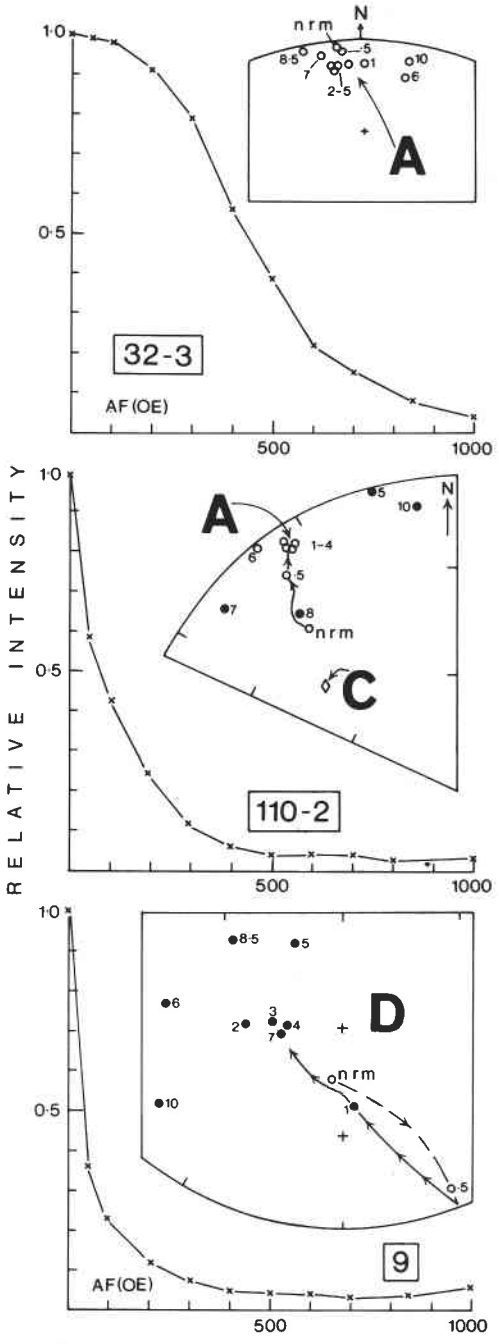


FIG. 5. Alternating field demagnetization curves and corresponding direction changes for the *A*, *C* and *D* magnetizations.

ilmenite is slightly altered to leucoxene, and the magnetite is extensively but not completely altered to an amorphous unidentified material. Corroded octahedrons of magnetite in the size range of about 20–50 μm occur without ilmenite lamellae, and the magnetite is again extensively altered. Pyrite occurs in minor amounts. In a few samples maghemite occurs, along with magnetite and its amorphous alteration product. Some flows have a reddish colour and in these the silicates are dusted with hematite and there are a few hematite grains.

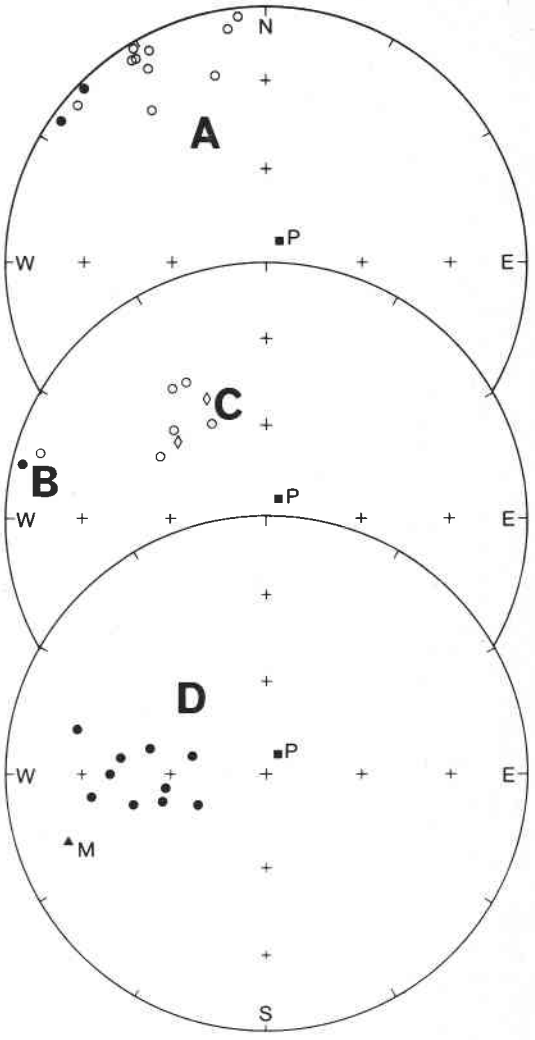


FIG. 6. Mean directions observed at each site. The treatments used are given in Table 2. *P* is the direction of the present geomagnetic field. Diamonds are directions obtained by vector subtraction. *M* is the direction of the Mackenzie diabase as observed in the sampling area (Irving *et al.* 1972).

Origin of Multicomponent Magnetization

Rocks of the Pearson Formation have a 'hard' *A* phase magnetization, a 'soft' *D* phase and a *C* phase of intermediate hardness. In addition to these clearly established magnetizations, there is a subordinate phase (*B*) that occurs only at two sites and that has not been fully investigated. The presence of both *A* and *C* phases (designated *A*+) in the same

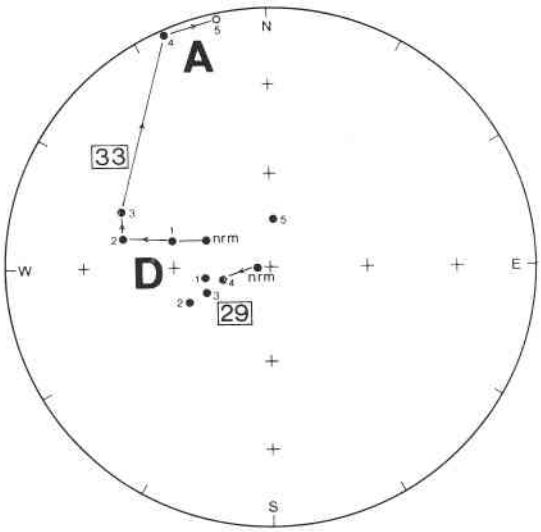


FIG. 7. Direction changes observed during thermal demagnetization in specimens with *D* magnetizations. Points are labelled in °C (× 10^2).

specimen is common. *A* and *D* phases occur together in only one specimen (Fig. 7). The magnetization directions at each site are listed in Table 2.

The *A* magnetization has high remanent coercive force and square-shouldered thermal decay curves. It is presumably carried by the unaltered magnetite observed in polished section. The results of Pullaiah *et al.* (1976) suggest that such magnetization would become unblocked in about 10^7 years at temperatures of about 400°C. This indicates that the *A* magnetization could have resisted heating associated with prehnite–pumpellyite facies metamorphism. Since the lavas do not exceed this metamorphic grade the *A* phase is probably the original magnetization, acquired at the time the lavas were extruded at approximately −1800 Ma. This argument is supported by all the known field relationships. The *A* direction is very different from that observed in the Et-Then Group (the pole positions PA and ET of Fig. 8 differ by more than three times the 95% error) and cannot be matched with any known secondary magnetization that has been observed from the region. The only known subsequent heating event is that associated with the massive Mackenzie intrusions (Fig. 1). Overprints identifiable with that event are found close to the intrusions (Irving *et al.* 1972), and sometimes at distances of the order of a kilometre (Evans and Bingham 1976; Irving and McGlynn 1978; poles TS and DS, Fig. 9). These overprints are readily removed at temperatures (usually about 300°C) much lower than the blocking temperatures of the *A* magnetization. Moreover, the *A* directions are over 60° different from the Mackenzie directions (Fig. 6). There is therefore no evidence that the *A* magnetization is secondary. The precision, before correction for tilt, is 21, compared with 15 after correction, but the change is not significant at $P = 0.05$ using Watson's test (Watson 1956); the dip contrasts are clearly too small to carry out a fold-test. Tilt corrections have no effect on the mean direction. The pole PA corresponding to the *A* magnetization is listed in Table 3 and plotted in Fig. 8.

The *B*, *C* and *D* magnetizations are probably secondary because they have lower blocking temperatures, and lower remanent coercivities, than the *A* magnetization, and because *B* and *C* can be tentatively identified with post-depositional events. The *D* magnetization presumably reflects a very mild reheating event because its blocking temperatures are mostly below 300°C. The pole (PD) corresponding to the *D* magnetization falls on the polar curve at about −1100 Ma (Fig. 9). Its origin is unknown. The pole (PB) for the *B* magnetization falls close to the pole for the Et-Then Group and could have been acquired during burial by the overlying Et-Then sandstones. The pole (PC) for the *C* magnetization falls in the Caribbean, and agrees with the poles for the overprints commonly found in rocks of the Coronation Geosyncline as first observed by McMurry *et al.* (1973). McGlynn *et al.* (1974) argue that these overprints were acquired during uplift and cooling followed the Hudsonian orogeny; it might be more precise to consider them to be caused by uplift and cooling following the orogeny in the Coronation Geosyncline (see below).

Coronation Loop

In Fig. 8 the poles obtained from the Canadian Shield in the interval −2200 to −1800 Ma are plotted. They are connected by a broad swath that is considered to describe the general motion of the pole relative to Laurentia. It is becoming obvious, however, as more detail becomes available, that the true path is more complicated. But it is not yet clear to what extent these complications are real irregularities in the path, or arise from local rotations in fault blocks, or are owing to the practical difficulties in getting accurate determinations of such old magnetizations. The polar sequence given by the stratigraphy of Table 1 is shown by arrows on Fig. 8. The Seton, Gibraltar and McLeod poles are in reasonable agreement, but there is a significant easterly trend. Whether this represents true polar movement, or relative rotations of blocks within the Athapuscow Aulacogen, is unknown. The pole (PA) derived from the *A* phase magnetiza-

TABLE 2. Results from Pearson Formation

Site[a]	Type	N^b	$M_n{}^c$	D^d (degree)	I^e (degree)	$(R)^f$	δ^g (degree)	Treatment[h]
8	*D*	3	82	262	+58	(2.98)	8	400 Oe
9	*D*	4	107	246	+66	(3.89)	15	300–500 Oe
10	*D*	5	158	284	+66	(4.79)	18	300–500 Oe
11	*D*	3	114	256	+56	(2.99)	5	300–500 Oe
12	*D*	4	92	283	+52	(3.97)	8	300–500 Oe
13	*D*	4	87	277	+43	(3.97)	7	300–500 Oe
27	*A*	4	62	332	−06	(3.93)	13	400 Oe, 500°C
28	*A*	5	1029	345	−26	(4.98)	6	400–500 Oe, 500°C
29	*D*	5	469	275	+46	(4.73)	21	300–1000 Oe
30	*A*	4	348	351	−08	(3.91)	14	400–1000 Oe, 500°C
32	*A*	4	134	354	−04	(3.97)	9	700–1000 Oe, 500 and 550°C
33	*D*	6	106	270	+39	(5.93)	10	500–600 Oe
34	*A*	4	62	314	+02	(3.92)	13	500 Oe
35	*D*	4	110	284	+26	(3.95)	11	300 Oe
40	*C*	5	10329	300	−51	(4.96)	9	400 Oe
42	*B*	4	1184	286	−10	(3.94)	12	300–500 Oe
43	*B*	4	808	283	+04	(3.73)	24	300–500 Oe
44	*C*	5	630	330	−55	(4.88)	14	nrm
44	*A*	4	—	310	−05	(3.98)	7	200 Oe
45	*C*	4	2234	324	−38	(3.94)	11	nrm
45	*A*	5	—	323	−27	(4.98)	6	300 Oe
107	*C*	2	8192	334	−46	(1.99)	9	
107	*A*	5		327	−06	(4.96)	8	300–400 Oe, 500°C
108	*C*	2	693	329	−38	(2.00)	5	500°C
108	*A*	3	—	329	−02	(2.99)	7	500°C
109	*D*	5	705	262	+32	(4.88)	14	200–500 Oe
109	*A*	3	—	305	+04	(2.95)	12	500°C
110	*A*	5	3423	329	−12	(4.98)	5	300–400 Oe, 500°C
110	*C*	3	—	312	−51	(2.94)	14	
111	*C*	2	3146	313	−50	(1.99)	9	nrm
111	*A*	6	—	327	−06	(5.95)	8	300–400 Oe

[a]The site numbers are accession numbers with no stratigraphic significance.
[b]*N*, number of separately oriented cores (2 specimens each).
[c]M_n, intensity of natural remanent magnetization (gauss × 10^{-6}).
[d]*D*, declination of the mean direction (degree).
[e]*I*, inclination of the mean direction (degree).
[f]*R*, resultant.
[g]δ, standard deviation (degree).
[h]In the right-hand column the stable end point cleaning fields and temperatures are given. These have been determined independently for each core. At site 107C the directions were obtained by subtracting the magnetization after cleaning in 400 Oe (31.8×10^3 A/m) af from the nrm. The directions of 110C were obtained in the same way except that the cleaning fields used were 200 and 300 Oe (15.9×10^3 and 23.9×10^3 A/m).
1 oersted = 79.577 47 A/m.

tion falls close to that for the Martin Formation (Evans and Bingham 1973), and these poles, together with those from the Douglas Formation, the Peninsular sill and the Takiyuak Formation, form a second group of poles about 20° further south. The pole for the Et-Then Group falls to the east near the equator, so that the polar path forms a broad loop. Other results, from the Churchill Structural Province, from Flin Flon (Park 1975) and from the Labrador Trough (Park 1977), are consistent with this loop, but the results from the Stark and Tochatwi Formations are not.

The Stark and Tochatwi Formations lie stratigraphically between the Douglas Peninsula and the Pearson Formations and their poles are situated in the central Pacific (Evans and Bingham 1976; Bingham and Evans 1976). The authors made several suggestions as to the reason for the peculiar position of these poles. The Stark and Tochatwi aberrant results were obtained from a restricted area west of Stark Lake (the TSTS sequence of Fig. 1) in which the strike is 60° different from the general strike of the rocks in the East Arm of the Great Slave Lake, and Bingham and Evans (1976) suggest that these rocks might have undergone rotations about a local axis. The TSTS region of Fig. 1 is certainly one of structural complexity with the rocks being intensely folded. Bingham and Evans (1976) calculate that the rotation necessary to bring the Stark strikes into conformity with those of the region would place the pole at 18°S, 083°W, close to where we now observe the Pearson PA pole to be. This in itself is evidence in favour of the rotation hypothesis because the Pearson conformably over-

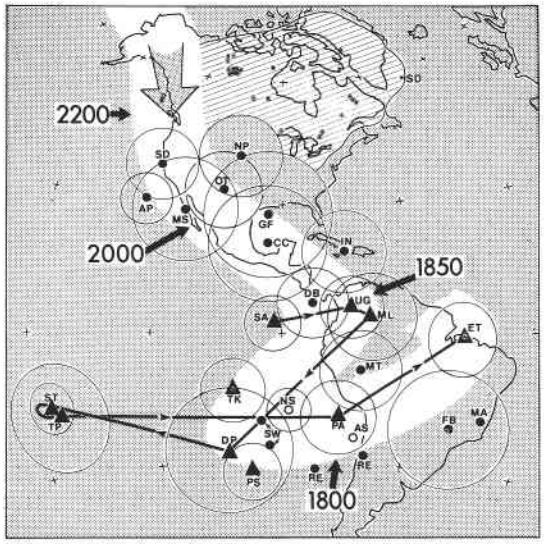


FIG. 8. Pole path for the interval −2200 to −1750 Ma. Assumed primary magnetizations (poles) from the Coronation Geosyncline are indicated as solid triangles, primary magnetizations from elsewhere in the Shield by solid dots and secondary magnetizations by open circles. Our best estimate of the polar path is shown as a broad swath. The pole path has been compiled from Irving and McGlynn (1976*b*), Roy and Lapointe (1976) and from the following sources:

Abbr.	Poles	Source	Age (Ma)
AP	Abitibi dikes primary magnetization	Larochelle 1966; Irving and Naldrett 1977	−2147
AS	Abitibi dikes secondary magnetization		
CC	Castignon Complex	Park 1977	−1830
DB	Dubawnt Group	Park *et al.* 1973	−1830
DP	Douglas Peninsula Formation primary magnetization	Irving and McGlynn 1978	
ET	Et-Then Group	Irving *et al.* 1972	
FB	Flin Flon greenstone belt *B* magnetization primary in Boundary Intrusions	Park 1975	−1800
GF	Gunflint Formation	Symons 1966	ca. −2200
IN	Indin dikes	McGlynn and Irving 1975	−2090
M	Mackenzie igneous events		−1250
MA	Menihek Formation, Labrador Trough, primary magnetization?	Park 1977	ca. −1800
ML	McLeod Formation	Reid 1972	
MS	Molson dikes	Ermanovics and Fahrig 1975	ca. −2000
MT	Martin Formation	Evans and Bingham 1973	ca. −1830
NP	Nipissing diabase primary magnetization	Roy and Lapointe 1976	−2150
NS	Nipissing diabase secondary magnetization	Roy and Lapointe 1976	
OT	Otish gabbro	Fahrig and Chown 1973	ca. −2000
PA	Pearson Formation *A* magnetization	Herein	−1800
PS	Peninsular sill	Irving and McGlynn 1978; R. K. Wanless, personal communication, 1977	−1800
RE	Two poles from the Retty peridotite, Labrador Trough	Park 1977	
SA	Seton Formation *A* magnetization	Irving and McGlynn 1978	−1870
SD	Scouri dikes corrected for closure of Atlantic	Beckman 1976	−2200 to −1900
ST	Stark Formation	Bingham and Evans 1976	
SW	Seward Formation *A* magnetization	Park 1977	
TK	Takiyuak Formation	Irving and McGlynn 1978	
TP	Tochatwi Formation	Evans and Bingham 1976	
UG	Upper Gibraltar Formation	Reid 1972	

TABLE 3. Summary of results

	B^a	N^b			k^c	$\alpha_{95}{}^d$ (degree)
A directions	12	52	329	−08	21	10
A poles	12	52	19S	077W	26	09
B directions	02	08	284	−03	64	32
B poles	02	08	05S	034W	227	17
C directions	07	23	321	−48	59	08
C poles	07	23	06N	077W	44	09
D directions	10	43	269	+49	26	10
D poles	10	43	27N	177E	25	10

[a] *B*, number of sites.
[b] *N*, total number of cores (2 specimens each).
[c] *k*, precision.
[d] α_{95}, radius of the cone of confidence ($P = 0.05$).

lies the Portage Inlet and Tochatwi Formations and it is unlikely that there is very much difference in age between the Tochatwi and Pearson rocks. Further evidence for the rotation idea has recently been obtained from the Takiyuak Formation (Irving and McGlynn 1978). The Takiyuak Formation is situated 400 km to the north outside the aulacogen and is autochthonous. It is the approximate stratigraphic equivalent of the Tochatwi (Hoffman 1973). Its pole is near the Douglas Peninsula pole and well to the east of the Stark and Tochatwi poles. Another possibility is that the polar path truly made a large swing westwards into the Pacific (Irving and McGlynn 1976*b*) or eastwards as far as Africa using the antipole of the poles (ST and TP) plotted in Fig. 8 (Roy and Lapointe 1976), meaning that all of Laurentia rotated about a nearby axis by about 60° between the deposition of the Stark and Pearson Formations. This could have happened, but, on the basis of present paleomagnetic evidence, the suggestion of Bingham and Evans (1976) that the Stark and Tochatwi poles are the result of local rotations, is the more reasonable one.

The polar paths of Figs. 8 and 9 are referred to as tracks 5 and 4. They correspond in time to the middle and late Aphebian and the Paleohelikian of Stockwell (1972). They are set out schematically in Fig. 10, which gives a summary polar path for Laurentia during the middle Proterozoic. The later part of track 5 and the early part of track 4 together comprise the Coronation loop. The loop is recorded by the magnetizations of the deposits of the Coronation Geosyncline and associated aulacogen. These deposits record the phases of a classical geosynclinal cycle for which Hoffman has provided a beautiful interpretation in terms of an ocean–continent plate interaction (Hoffman 1973). The Coronation loop appears, in part at least, to be the paleomagnetic signature of this cycle. The rate of motion relative to the pole, that is the rate of change of latitude, is about 5 cm year^{-1}. The major deformation phase occurs during and after the deposition of the Great Slave Supergroup, but before the deposition of the Et-Then Group, so the bend or hairpin of the Coronation loop corresponds to that deformational phase. During this deformational phase, the Coronation Geosyncline was in latitudes of about 25° (Seton Formation), moved to equatorial latitudes (Christie Bay Group) and then back to a latitude of about 25° (Et-Then Group). Thus the major deformation phase of the Coronation Geosyncline corresponds to a large change in direction of the motions of the Laurentian plate as recorded paleomagnetically between the Pearson and the Et-Then.

The orogenic phase of the Coronation Geosyncline, which is dated about −1800 Ma, appears to post-date certain Hudsonian events in the central part of the Churchill Province, because the Dubawnt Group, which post-dates the major deformation of the central part of the Churchill Province, has a radiometric age and pole position that agrees with those of the Seton Formation in the lower part of the Great Slave Supergroup. The age of the Seton corresponds to ages of syntectonic granitic rocks (−1870 Ma), and are younger than the upper Hurwitz volcanic rocks of the Churchill Province (−1910 Ma, Wanless and Eade 1975). This suggests that the orogeny, and much of the depositional phase of the Coronation Geosyncline, are younger than the orogenic and depositional phases of the central part of the Churchill Province. It is unwise therefore to use the term Hudsonian orogeny to describe the orogeny in the Coronation Geosyncline.

Possible Aphebian–Helikian Boundary Stratotype

The Coronation loop could have some interest-

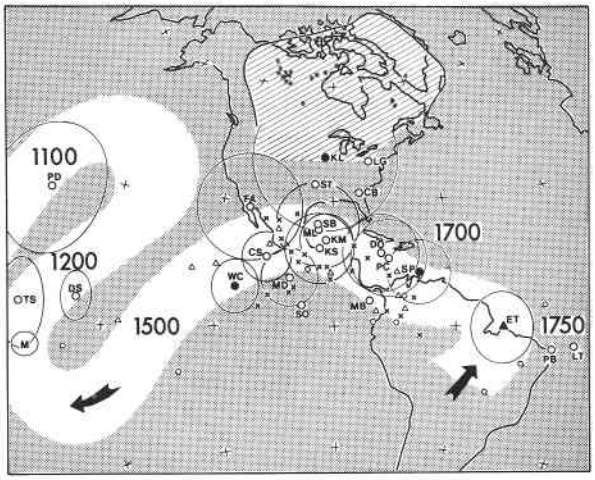


FIG. 9. Pole path for the interval about −1750 Ma to about −1100 Ma. The individual poles for the interval about −1700 to −1600 Ma are plotted, including overprints. The polar path between −1500 and −1200 Ma is shown as a simple loop that is the form commonly accepted by most reviewers. The more complex path recently suggested by Irving *et al.* (1977) is shown in Fig. 10. Nonacho Group secondary magnetizations are denoted by open triangles (McGlynn *et al.* 1974), the Big Spruce overprints by open circles (Irving and McGlynn 1976*a*) and the Greenland Nagssugtoquidian dikes secondary magnetizations by crosses (Morgan 1976). The individual poles are as follows:

Abbr.	Poles	Source
CB	Charlton Bay Formation overprint?	Reid 1972
CS	Cape Smith overprint?	Fujiwara and Schwarz 1975
DO	Dogrib dikes overprint	McGlynn and Irving 1975
DS	Douglas Peninsula Formation overprint	Irving and McGlynn 1978
FA	Flin Flon *A*	Park 1975
KL	Kaminak lamprophyre	Christie *et al.* 1975
KM	Kaminak metadiabase dikes	Christie *et al.* 1975
KS	Kahochella Group overprint	McMurry *et al.* 1973
LG	Gibraltar Formation lower part	Reid 1972
LT	Dubawnt Group lapilli tuff, overprint	Park *et al.* 1973
MB	Menihek Formation overprint	Park 1977
MD	Melville – Daly Bay metamorphic rocks	Park 1973
ML	McLean Formation overprint?	Reid 1972
PB	Pearson Formation *B*	Herein
PC	Pearson Formation *C*	Herein
PD	Pearson Formation *D*	Herein
SB	Seton Formation overprint?	Reid 1972
SO	Seward Formation overprint	Park 1977
SP	Sparrow dikes	McGlynn *et al.* 1974
TS	Tochatwi Formation overprint	Evans and Bingham 1976
WC	Western Channel diabase	Irving *et al.* 1972

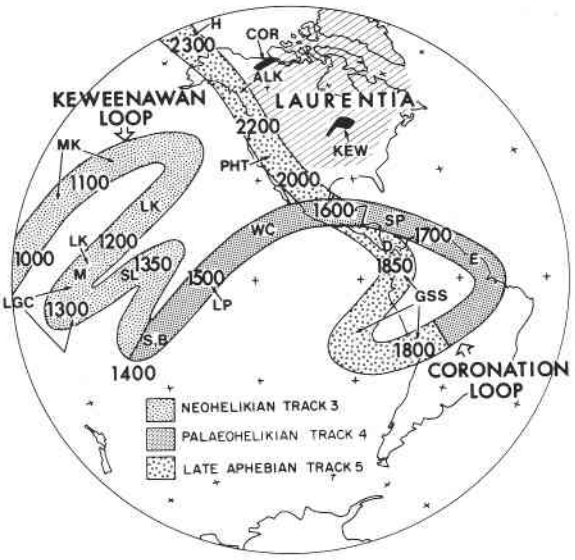


FIG. 10. Summary polar path for Middle Proterozoic of Laurentia. The path is calibrated in millions of years and has been compiled mainly from Figs. 8 and 9 and from Du Bois (1962), Books (1972), Robertson and Fahrig (1971), Irving and McGlynn (1976*b*), Roy and Lapointe (1976) and Irving *et al.* (1977). The rock units from which the more important results have been obtained are as follows from oldest to youngest: H, Huronian Supergroup; ALK, Otto Stock and Big Spruce alkaline complexes; PHT, post-Huronian tholeiitic intrusions; D, Dubawnt redbeds and volcanic rocks; GSS, Great Slave Supergroup; E, Et-Then Group; WC, Western Channel diabase; LP, Labrador anorthosite and adamellite plutons north of Grenville Front; SB, Sibley Group and Belt Supergroup; SL, Seal Lake Group; LGC, lower formations of Grand Canyon Supergroup; M, Mackenzie diabase; LK, lower Keweenawan Group; MK, middle Keweenawan Group. References to all these studies are given in the reviews cited above and in the Earth Physics Branch catalogue of Precambrian paleomagnetic results (Irving and Hastie 1975).

ing stratigraphic implications. The changes in magnetic directions it signifies mark the transition from the late Aphebian track 5 to the Neohelikian track 4. This change is observed to occur between the Et-Then Group and the underlying Pearson Formation. We suggest, tentatively, that this may serve as a useful physically recognizable boundary between the Aphebian and Helikian. The type location is the East Arm of the Great Slave Lake, and the pole change corresponding to the observed magnetic change is as described in Fig. 10. The current best estimate of the time at which this change occurred is −1800 Ma. The Et-Then Group contains thin basic volcanic members (Hoffman 1968), and hence, like the Pearson Formation, could possibly be dated, so that in principle the boundary could be more accurately fixed in time.

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